

Below is a **practical “all major database types” list with when to use vs when NOT to use** (real-world software development).

✓ Table format as you requested.

✓ **Database Types – When to Use vs When NOT to Use (Master Table)**

Database Type	Common Examples	Best For (When to Use)	Avoid / Don't Use When
Relational (RDBMS / SQL)	PostgreSQL, MySQL, Oracle, SQL Server	Banking, ERP, CRM, transactions, strict schema, joins, strong data consistency	Highly unstructured data, very high write scale without sharding, flexible schema needs
Distributed SQL (NewSQL)	Google Spanner, CockroachDB, YugabyteDB, TiDB	Global apps needing SQL + strong consistency + scalability	Simple apps (too complex), low budget, small workloads
NoSQL – Key-Value Store	Redis, DynamoDB, Riak	Extremely fast lookups, sessions, tokens, cache, leaderboards	Complex queries, joins, reporting/analytics
NoSQL – Document Database	MongoDB, CouchDB, Firestore	JSON-based apps, flexible schema, mobile/web apps, product catalogs	Complex join-heavy relational design, strict ACID transactions across many tables
NoSQL – Column-Family / Wide-Column	Cassandra, HBase, ScyllaDB	Massive scale, high write throughput, time-series-like records, IoT	Ad-hoc querying, joins, strong consistency requirements
NoSQL – Graph Database	Neo4j, JanusGraph, Amazon Neptune	Social networks, recommendations, fraud detection, relationship-heavy queries	Simple CRUD apps; if relations are minimal
Time-Series Database (TSDB)	InfluxDB, TimescaleDB, Prometheus TSDB	Monitoring metrics, sensor data, logs metrics, time-based aggregation	Non-time related business data (users/orders)
Data Warehouse (Analytics DB)	Snowflake, BigQuery, Redshift, Teradata	BI reports, dashboards, huge analytics queries, OLAP	Real-time transactional systems (slow/expensive per transaction)
OLAP / Columnar Analytics DB	ClickHouse, Apache Druid, Apache Pinot	Ultra-fast analytics on large datasets, event analytics	Heavy transactional workloads (OLTP)

Search Engine DB	Elasticsearch, OpenSearch, Solr	Full-text search, relevance ranking, log search	Primary transactional DB (no strong ACID, risk of data loss)
In-Memory Database	Redis, Memcached, SAP HANA	Ultra-low latency caching, session store, realtime counters	Permanent storage (unless persistence is enabled properly)
Embedded Database	SQLite, LevelDB, RocksDB	Mobile apps, edge devices, local storage, lightweight apps	Multi-user high concurrency server apps
Multi-Model Database	ArangoDB, OrientDB, Cosmos DB	Mixed workloads (document + graph + KV)	If one model dominates → best use specialized DB
Object-Oriented Database (OODBMS)	ObjectDB, db4o (legacy)	Niche apps with object persistence	Modern enterprise systems (rare ecosystem/support)
Hierarchical Database	IBM IMS	Legacy banking/mainframe systems	New projects (hard to maintain & evolve)
Network Database	IDMS	Legacy systems	New systems (obsolete design style)
Ledger Database	Amazon QLDB, Hyperledger	Immutable audit trails, tamper-resistant records	General CRUD apps; fast reporting + joins
Blockchain as DB	Ethereum, Hyperledger Fabric	Decentralized trust, smart contracts	Performance-sensitive apps; simple internal apps
Vector Database (AI DB)	Pinecone, Milvus, Weaviate, Chroma	AI search, embeddings, RAG, semantic search	Traditional transactional business data
Spatial / GIS Database	PostGIS (PostgreSQL), Oracle Spatial	Maps, geofencing, routing, nearest location search	Apps without geospatial requirement
Real-Time Database	Firebase Realtime DB	Realtime sync apps (chat, live dashboards)	Complex queries, reporting, heavy analytics
Event Store Database	EventStoreDB, Kafka + storage	Event sourcing, audit logs, replayable state	Simple CRUD systems
Streaming Platform (DB-like use)	Apache Kafka, Pulsar	Real-time pipelines, pub-sub, event-driven microservices	Storing business state permanently (Kafka ≠ DB)

Columnar RDBMS	Vertica, ClickHouse, Redshift	Analytics, scan-heavy queries	Frequent updates/deletes (not optimal)
Flat File Storage (Not DB but used as one)	CSV, JSON, XML	Simple exchange/import/export, temporary storage	Multi-user concurrent apps; strict reliability needs
Cloud Native DB (Managed)	Aurora, DynamoDB, Cosmos DB, Spanner	Faster scaling, managed backups, HA/DR included	On-prem required; strict cost control without monitoring

✔ Quick Decision Cheat Sheet (Most Common Choices)

Use Case	Best Database
Banking / payments / orders	PostgreSQL / MySQL / Oracle
High scale sessions / cache	Redis
Flexible JSON app backend	MongoDB
Social network / recommendation engine	Neo4j
Monitoring & metrics	InfluxDB / Prometheus / TimescaleDB
BI dashboards (company reporting)	Snowflake / BigQuery / Redshift
Full-text search	Elasticsearch / OpenSearch
AI semantic search (RAG)	Pinecone / Milvus / Weaviate
Event-driven microservices	Kafka (+ Postgres for state)

If you want, I can also provide:

- ✔ **“All databases list by name (100+)”** in a table
- ✔ **Database vs Database comparison matrix** (PostgreSQL vs MongoDB vs Cassandra vs Redis etc.)
- ✔ **Which DB to use per project type** (E-Commerce / Banking / Social media / AI chatbot / IoT)

below is a **100+ databases list by name** with **When to use vs When NOT to use** in a single practical table.

Note: This list covers **RDBMS, NoSQL, Graph, Vector, TSDB, Search, Analytics/OLAP, Embedded, Ledger, Streaming DB-like systems**, etc.

✔ All Databases List (100+) — When to Use vs When NOT to Use (Table)

Database Name	Category	When to Use (Best For)	When NOT to Use (Avoid)
PostgreSQL	Relational (SQL)	Best general-purpose OLTP, ACID, joins, extensions	Extreme scale without sharding strategy
MySQL	Relational (SQL)	Web apps, simple OLTP, read-heavy	Complex analytics, heavy concurrency at scale
MariaDB	Relational (SQL)	MySQL compatible systems	If strict enterprise Oracle-like features needed
Oracle Database	Relational (SQL)	Enterprise banking/ERP, HA, compliance	Low budget projects
Microsoft SQL Server	Relational (SQL)	Enterprise apps, MS ecosystem	Linux-native lightweight deployments
IBM Db2	Relational (SQL)	Banking enterprise, mainframes	Small/medium apps (overkill)
SAP HANA	In-memory / Hybrid	Real-time analytics + OLTP	Cost-sensitive apps
SQLite	Embedded SQL	Mobile/local storage, lightweight apps	High concurrency multi-user server systems
Microsoft Access	Desktop DB	Very small office apps	Production APIs, scalable apps
Amazon Aurora	Managed SQL	HA managed MySQL/Postgres	Vendor lock-in concerns
Amazon RDS	Managed SQL	Managed relational DB workloads	Ultra-high scale without tuning

Google Cloud SQL	Managed SQL	Standard SQL in GCP	Massive distributed global OLTP
Azure SQL Database	Managed SQL	SQL Server cloud SaaS	Very complex cross-region low latency
YugabyteDB	Distributed SQL (NewSQL)	Globally distributed SQL workloads	Small apps; operational complexity
CockroachDB	Distributed SQL (NewSQL)	Strong consistency + horizontal scale	Low-latency single region apps (Postgres better)
Google Spanner	Distributed SQL	Global consistency & scale	Small projects; cost
TiDB	Distributed SQL	MySQL compatible scale-out	Small/simple apps
VoltDB	NewSQL	Ultra-fast real-time transactions	General apps; complex analytics
SingleStore	NewSQL / Analytics	HTAP (trans + analytics)	Small teams (learning curve/cost)
Redis	Key-Value / In-memory	Cache, sessions, queues, counters	As primary DB for business state
Memcached	Cache	Simple caching	Persistent storage
Amazon DynamoDB	Key-Value / NoSQL	Serverless scale KV workloads	Complex join queries/reporting
Riak KV	Key-Value NoSQL	Fault tolerant KV	Modern apps (less popular ecosystem)
Aerospike	KV / NoSQL	Extreme low latency large scale	If you require complex relational queries
LevelDB	Embedded KV	Local fast KV	Querying/analytics needs
RocksDB	Embedded KV	High-performance storage engine	Enterprise SQL / BI

LMDB	Embedded KV	Fast transactional embedded KV	Distributed/clustered apps
Berkeley DB	Embedded KV	Old stable embedded database	Modern distributed systems
MongoDB	Document NoSQL	Flexible JSON apps, catalogs, content	Heavy join/transactional accounting style
CouchDB	Document NoSQL	Offline-first sync apps	High-scale analytics
Couchbase	Document + KV	Low latency doc store	Heavy SQL analytics/reporting
RavenDB	Document NoSQL	ACID documents, .NET friendly	If you need standard SQL ecosystem
Firebase Firestore	Document DB	Mobile backend, realtime sync	Complex reporting/joins
Firebase Realtime DB	Realtime DB	Live sync apps (chat/live tracking)	Complex queries & analytics
MarkLogic	Document / Enterprise	Search + doc + enterprise	Low budget
Amazon DocumentDB	Document DB	Mongo-compatible managed AWS	If you need full MongoDB feature parity
Apache Cassandra	Wide-column NoSQL	Very high writes, IoT scale	Joins, complex querying
ScyllaDB	Wide-column	Cassandra-compatible faster engine	Join-heavy systems
Apache HBase	Wide-column	Hadoop ecosystem huge storage	General app DB
Google Bigtable	Wide-column	Massive scale time-series-like	Small/medium apps
Hypertable	Wide-column	Rare legacy alternative	Modern projects
Neo4j	Graph DB	Relationship queries, recommendation	Simple CRUD apps

Amazon Neptune	Graph DB	Managed graph in AWS	If graph queries are not core
JanusGraph	Graph DB	Graph layer over storage engines	Small teams/complex ops
TigerGraph	Graph DB	Large-scale graph analytics	Small apps
Dgraph	Graph DB	Distributed graph queries	Standard OLTP apps
OrientDB	Multi-model Graph	Mixed document+graph	Strict SQL apps
ArangoDB	Multi-model	Document + graph + KV	If one model dominates (use specialized DB)
Cosmos DB	Multi-model cloud	Global distribution, multiple APIs	Lock-in + cost without planning
InfluxDB	Time-series DB	Metrics/IoT time queries	Business transactions
TimescaleDB	Time-series (Postgres ext)	TS + SQL + Postgres power	If you need pure NoSQL ingestion scale
Prometheus TSDB	Metrics TSDB	Monitoring metrics	Business DB
OpenTSDB	TSDB	Time-series at scale	Modern app DB
QuestDB	TSDB	High-frequency time series	User/order transactions
TDengine	TSDB	Industrial IoT	Ecommerce/banking OLTP
VictoriaMetrics	TSDB	Metrics storage at scale	OLTP
Elasticsearch	Search engine	Full-text search, logs search	Primary OLTP database
OpenSearch	Search engine	Elasticsearch alternative	Strong ACID systems
Apache Solr	Search engine	Enterprise search	Transaction DB

Meilisearch	Search engine	App search, fast setup	Analytics OLTP
Typesense	Search engine	Instant search	Primary DB
ClickHouse	OLAP columnar	Event analytics very fast	Frequent updates/deletes
Apache Druid	OLAP realtime	Real-time analytics dashboards	OLTP systems
Apache Pinot	OLAP realtime	User-facing analytics	Financial transactions
BigQuery	Data Warehouse	Serverless analytics	High-frequency OLTP
Snowflake	Data Warehouse	BI, ETL, analytics	OLTP
Amazon Redshift	Warehouse	AWS BI warehouse	OLTP & frequent small updates
Azure Synapse	Warehouse	Microsoft analytics	OLTP
Vertica	Columnar analytics	High performance OLAP	CRUD systems
Greenplum	MPP warehouse	On-prem analytics	OLTP
Teradata	Enterprise warehouse	Huge BI organizations	Small businesses
DuckDB	Embedded analytics	Local OLAP, data science	Multi-user OLTP
Apache Hive	SQL on Hadoop	Batch analytics	Realtime querying
Apache Impala	SQL analytics	Hadoop interactive analytics	Transactional apps
Trino	SQL query engine	Query many systems	Not a storage DB
Presto	SQL query engine	Interactive analytics	Not a DB
Apache Spark SQL	Compute engine	ETL analytics	Not primary DB
Microsoft Azure Data Explorer (Kusto)	Analytics DB	Logs & telemetry queries	OLTP

Grafana Loki	Log DB	Logs + labels	OLTP
Splunk (storage)	Log analytics	Security logs and search	Primary DB storage
Apache Kafka	Streaming platform	Event streaming, pub/sub	Not a DB for state
Apache Pulsar	Streaming	Streaming at scale	Transaction state
Redpanda	Kafka-compatible	Streaming without JVM	Not DB replacement
EventStoreDB	Event DB	Event sourcing systems	Simple CRUD apps
NATS JetStream	Messaging/stream	Real-time events	DB replacement
RabbitMQ (storage queues)	Messaging	Reliable queues	Not a DB
Milvus	Vector DB	Embeddings, RAG search	OLTP data
Pinecone	Vector DB	Managed semantic search	General transactional apps
Weaviate	Vector DB	RAG + metadata filtering	Accounting transactions
Chroma	Vector DB	Local dev RAG	Enterprise HA
Qdrant	Vector DB	High-perf vector similarity	OLTP
FAISS (library)	Vector index	Local similarity search	Not persistent DB
Vespa	Search + vector	Large scale search + ML	Simple CRUD
pgvector (Postgres ext)	Vector	Keep AI search inside Postgres	Very high vector-only scale
Amazon OpenSearch Vector	Vector search	AWS vector search	Strict OLTP
Oracle Spatial	Spatial DB	GIS enterprise	Apps without GIS needs
PostGIS (Postgres ext)	Spatial	Maps, distance queries	No location-based needs

MongoDB Geo	Spatial feature	Geo queries inside MongoDB	Complex GIS systems
Redis GEO	Spatial	Geo lookup fast	Advanced GIS
Amazon Timestream	TSDB	Managed time-series	OLTP
Amazon QLDB	Ledger DB	Immutable audit trails	Standard CRUD
Hyperledger Fabric	Blockchain ledger	Enterprise blockchain	Performance-sensitive apps
Ethereum (smart contracts)	Blockchain	Decentralized apps	Normal apps requiring speed
BigchainDB	Blockchain-like DB	Immutable records	Modern enterprise standards
Datomic	Immutable DB	Audit/history based systems	Typical CRUD apps
FoundationDB	Distributed KV	Strong foundation layer	If SQL is needed directly
Apache Accumulo	Wide-column secure	Govt/secure apps	Normal SaaS CRUD
Realm	Mobile DB	Offline-first mobile	Complex server reporting
ObjectDB	Object DB	Java object persistence	Modern microservices
InterSystems IRIS	Multi-model	Healthcare/enterprise	Simple apps
Informix	RDBMS	Legacy enterprise	New builds
Sybase ASE	RDBMS	Legacy enterprise systems	New builds
Firebird	Relational DB	Small/embedded relational	Large distributed scale
MonetDB	Columnar DB	Analytics and research	OLTP apps
Apache Derby	Embedded SQL	Java embedded DB	High concurrency server
H2 Database	Embedded SQL	Dev/test, lightweight	Production at scale

HyperSQL (HSQLDB)	Embedded SQL	Dev/test	Heavy production
MariaDB ColumnStore	Columnar	OLAP on MariaDB	High update workloads
Apache Kudu	Columnar storage	Analytics in Hadoop	Complex OLTP
RocksDB (as engine in systems)	Embedded	Storage layer for DBs	Not for BI directly
